SANJAI S

Full-Stack Software Engineer

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PROFILE

Full-stack software engineer with hands-on experience building production systems in Next.js, React, TypeScript, Node.js, and PostgreSQL, plus a strong foundation in Java, Python, SQL, and data structures & algorithms. Comfortable owning a feature end-to-end — schema design, REST APIs, auth/authorization, and deployment — with a track record of shipping enterprise-grade features (RBAC, RLS, audit trails) during a live production internship.

SKILLS

Languages: Java, Python, SQL, TypeScript, JavaScript
Frontend: React, Next.js, Tailwind CSS
Backend: Node.js, Express.js, Spring Boot, REST APIs
Databases: PostgreSQL, MySQL, Supabase, Neon
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
DevOps & Tools: Git & GitHub, Docker, Postman
Cloud & Deployment: AWS, Vercel, Render

EXPERIENCE

Software Development Intern — Tamil Nadu National Highways Department
Chennai, TN | Aug 2026 – Sep 2026

Highways Department Management System *| Next.js, React, TypeScript, PostgreSQL, Supabase, Tailwind CSS*

- Built an enterprise full-stack platform to manage highway corridors, infrastructure projects, maintenance contracts, legal cases, fee bills, and departmental staff across six production modules.
- Implemented REST API route handlers and relational PostgreSQL schemas with strongly typed data models, shipping 52 incremental database migrations without downtime.
- Implemented enterprise-grade authentication and authorization: TOTP-based 2FA, hierarchical RBAC, PBAC jurisdiction scoping, PostgreSQL Row-Level Security (RLS), and database-backed rate limiting.
- Built reusable progressive forms, dashboards, search/filtering, soft-delete workflows, and Excel export functionality reused across all six modules.
- Developed an immutable audit trail capturing record creation, modification, deletion, restoration, and role changes, including field-level diffs and historical snapshots.

Inward Tapal Management System *| Node.js, Express.js, PostgreSQL, JavaScript, Supabase*

- Developed a full-stack inward letter and document tracking system covering registration, assignment, workflow progression, and archival of official communications across departmental offices.
- Designed and implemented RESTful APIs for authentication, CRUD operations, search/filtering, workflow management, and system configuration.
- Implemented Role-Based Access Control (RBAC) for multiple administrative roles, with date-lock mechanisms and authorization controls to prevent unauthorized modifications.
- Designed a PostgreSQL schema with dedicated tables for users, offices, tapal records, workflow history, audit logs, and system settings, using triggers and constraints for data integrity.
- Optimized database search using PostgreSQL pg_trgm text matching and connection pooling; deployed across Supabase, Vercel, and Linux servers via PM2.

PROJECTS

IPL Performance Analysis Dashboard *| Python, Pandas, Streamlit, Plotly, Three.js*

- Built an interactive web app processing 301,000+ ball-by-ball deliveries across 1,169 IPL matches (2008–2025) using Python and Pandas.
- Engineered interactive 3D pitch maps and wagon-wheel visualizations with Three.js/WebGL, rendering up to 500 ball trajectories per view.
- Built reusable data-processing and analytical modules for phase-wise statistics, batter–bowler matchups, strike rates, and economy metrics.
- Optimized performance for a 44.5 MB dataset using vectorized Pandas operations and Streamlit caching, and integrated data processing, visualization, and UI into a single deployable end-to-end application.

EDUCATION

B.E. Computer Science Engineering (Artificial Intelligence & Machine Learning) 2023 – 2027
Velalar College of Engineering & Technology, Thindal, Erode CGPA: 7.01 / 10

CERTIFICATIONS

- **HackerRank SQL (Advanced)** — Skill certification demonstrating expertise in joins, CTEs, window functions, subqueries, aggregation, and query optimization (July 2026).
- **Deploying Gen AI with AWS** — Workshop, Sri Eshwar College of Engineering (Feb 2025).

ACHIEVEMENTS

- **1st Prize** — Smart Bus Optimization Technical Competition (2025) | Finalist — VIT Hackathon (2026)
- **Qualified** — College-Level Internal Hackathon for Smart India Hackathon (SIH) 2025.